Galactic Astronomy

2024-2025

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Inherited from:

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EXCELENCIA MARÍA DE MAEZTU

UNIVERSITAT DE BARCELONA

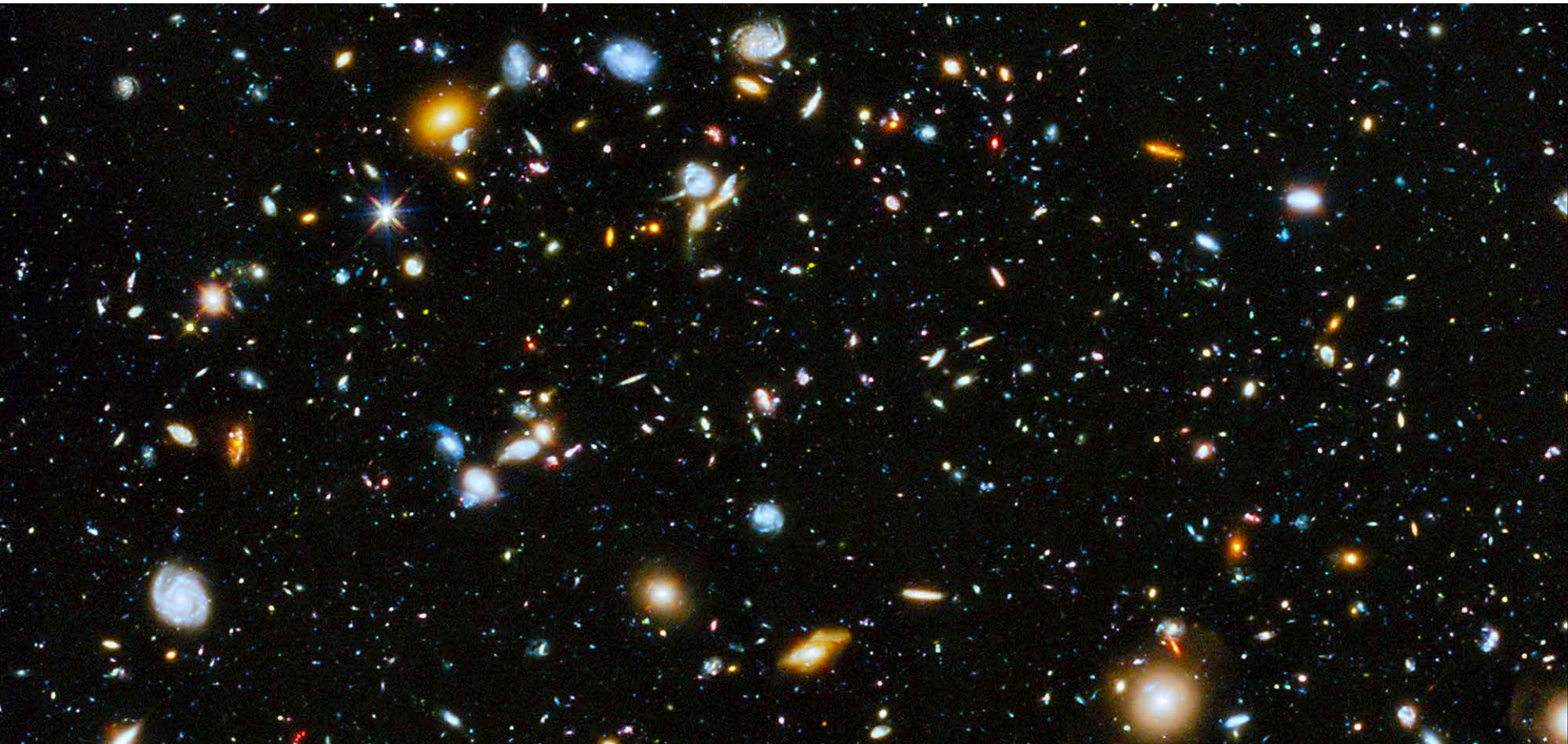
U Gaia B

How do galaxies form and evolve?

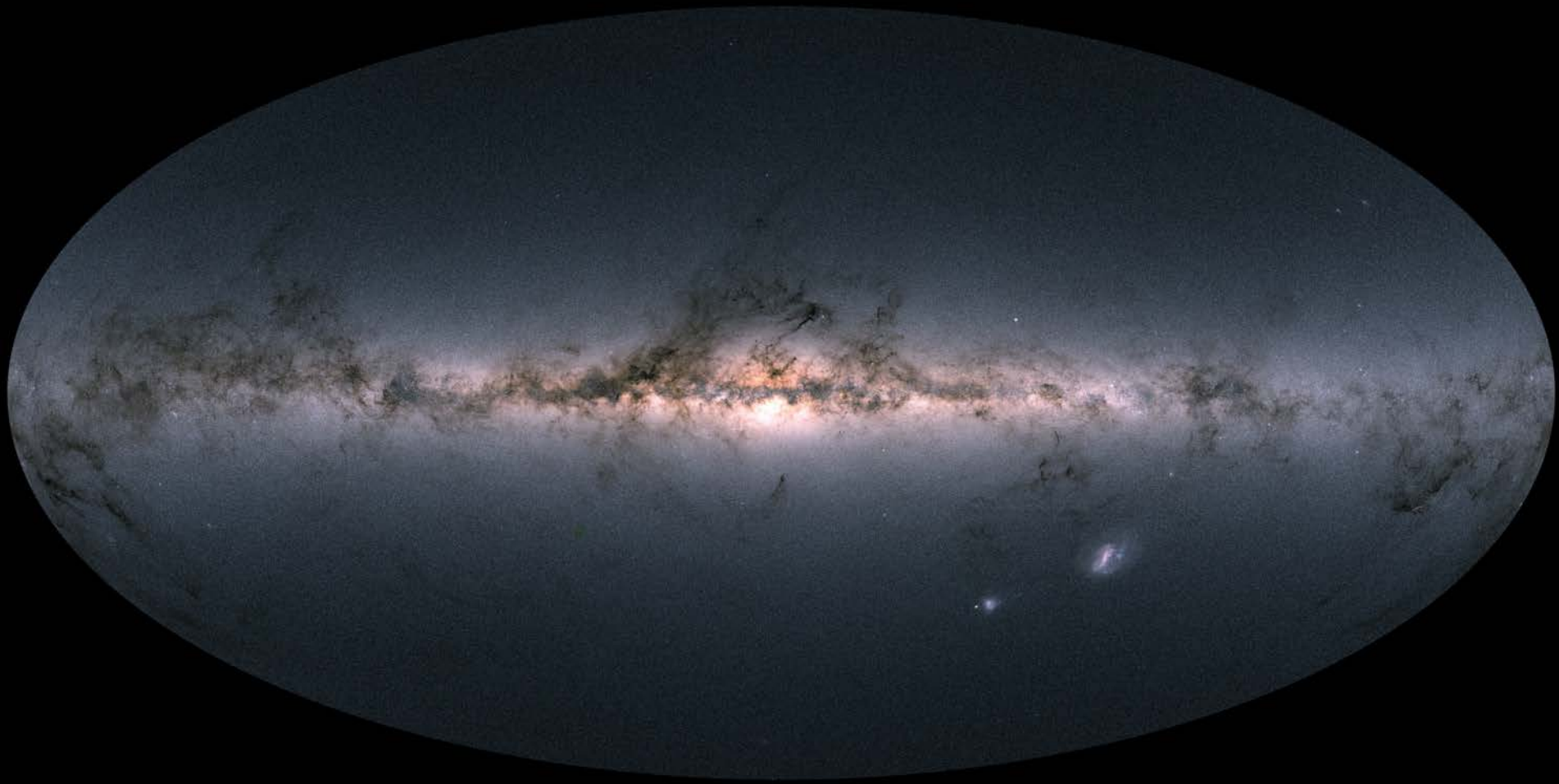
One of the key questions in modern astrophysics:

- A Science Vision for European Astronomy, ASTRONET, 2007
- Science Vision and Infrastructure Roadmap, ASTRONET, 2022 *(DRAFT)*

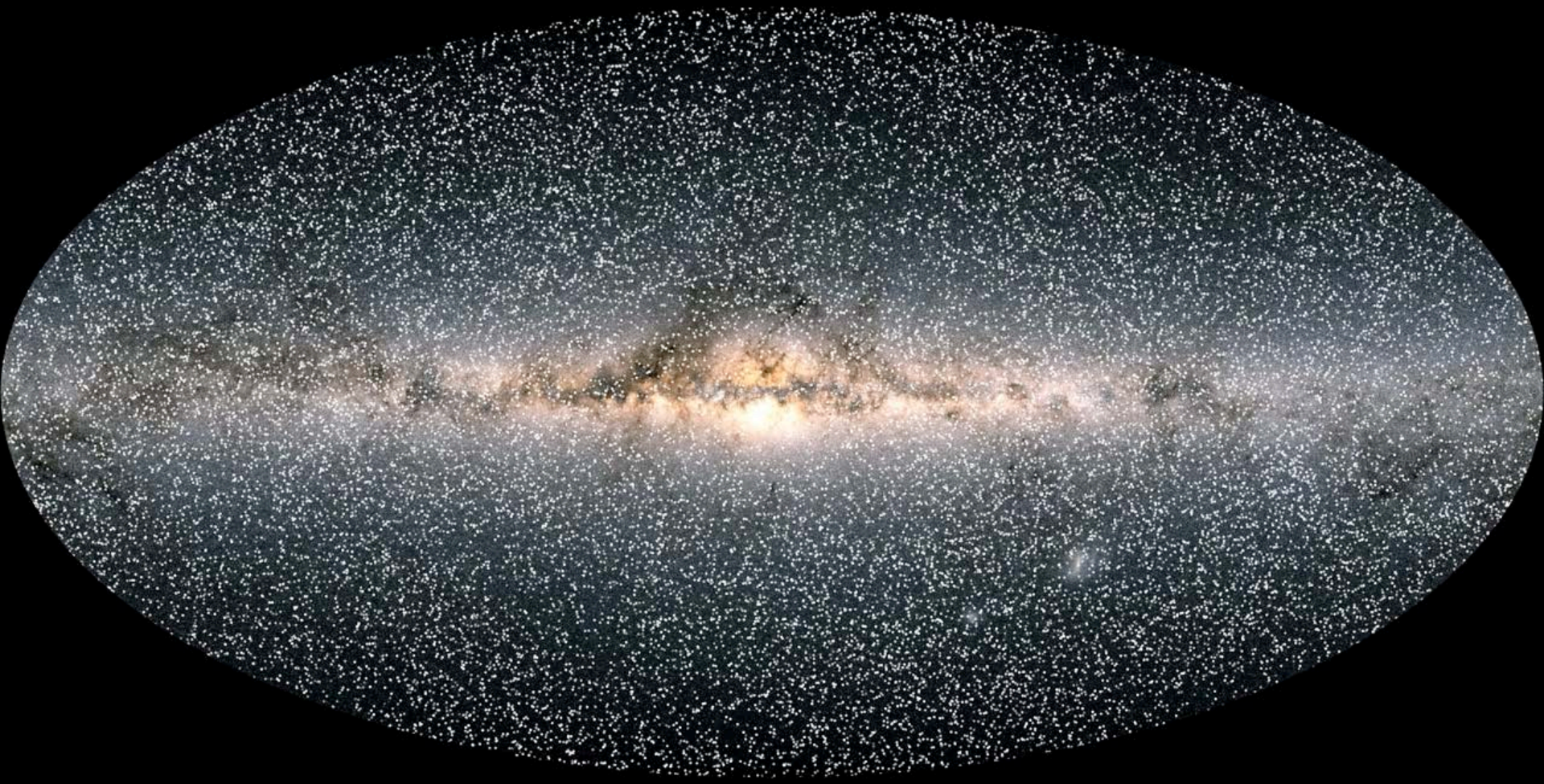
Hubble Ultra Deep Field



Gaia DR2



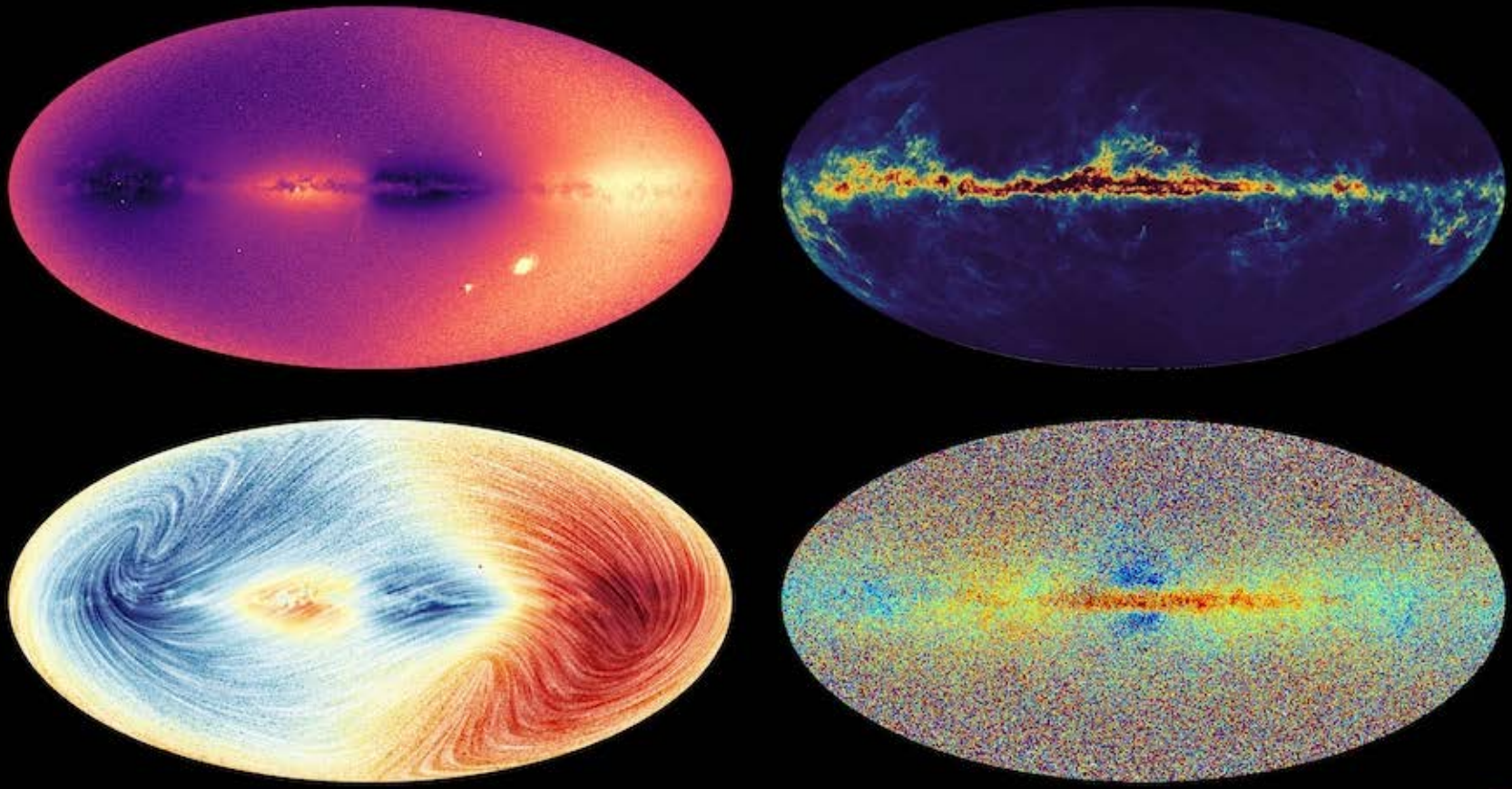
Gaia EDR3



Gaia DPAC

esa

GAIA: EXPLORING THE MULTI-DIMENSIONAL MILKY WAY

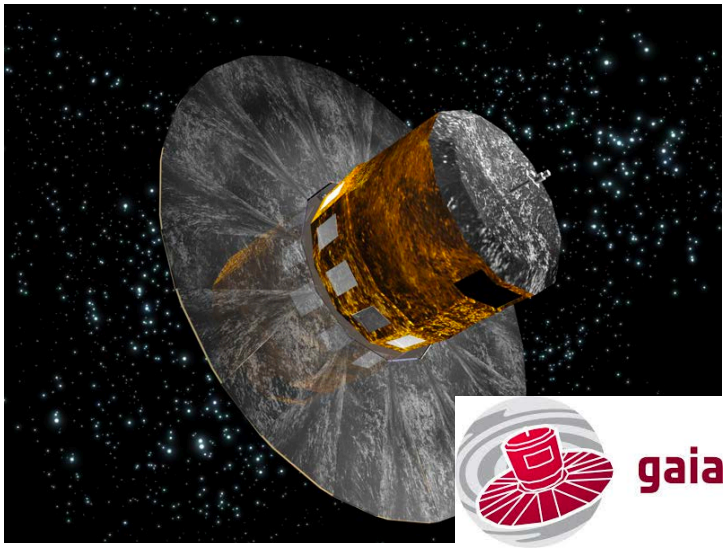


www.esa.int

Credit: ESA/Gaia/DPAC

Measurements: Gaia mission

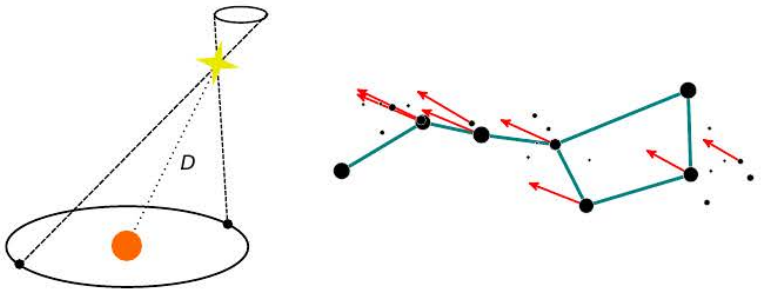


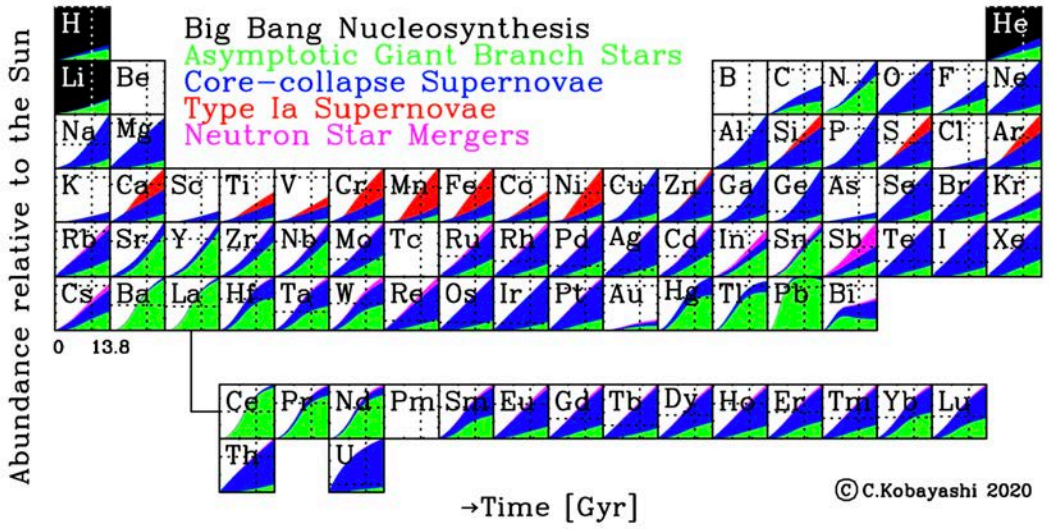


December 19th, 2013
10:12 CET

How did our galaxy and its components form?

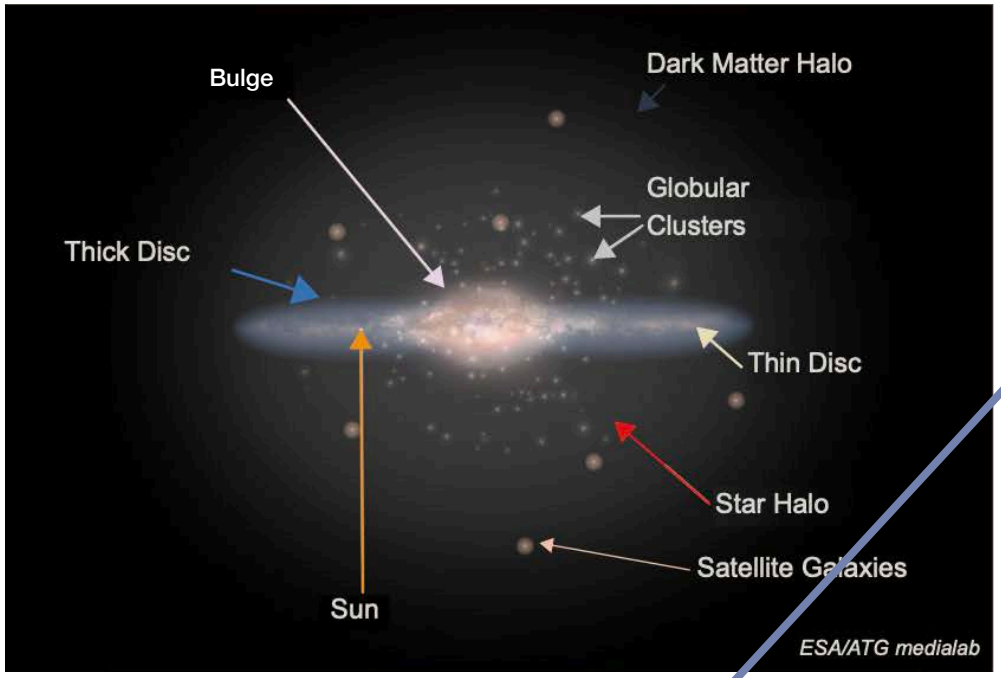
Astronomical measurements in Galactic Astronomy

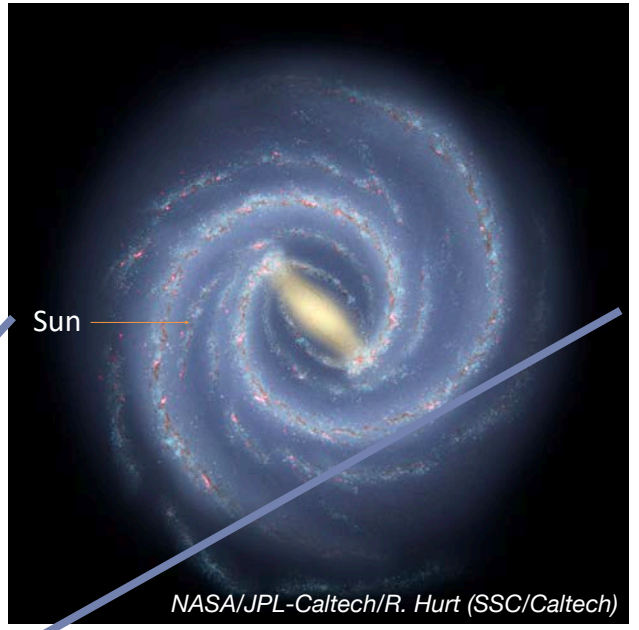


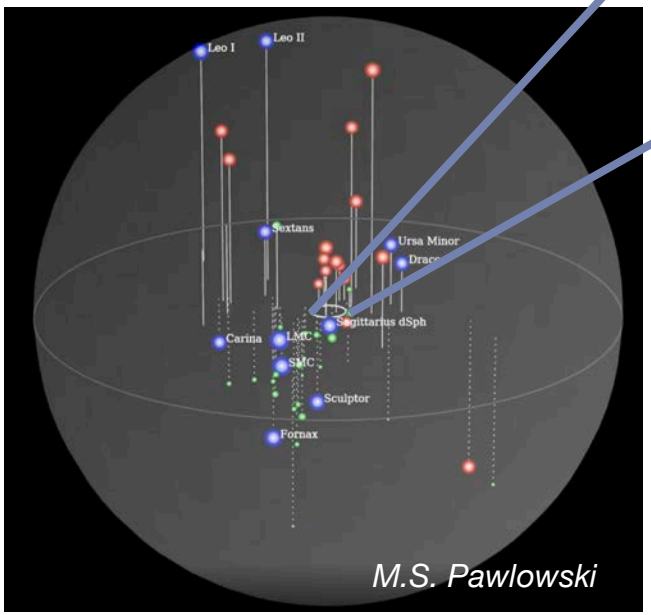


- Galaxy gravitational potential (mass distribution)
- History and evolution

What we think the MW looks like







5 themes:

- Galactic dynamics
- Astronomical measurements
- Statistical galactic astronomy
- Galactic kinematics
- Chemical evolution of the Milky Way

PROGRAMA

1. Introduction TERESA
   16 set
   - 1.1. Galaxies and their role in the Universe
   - 1.2. Galactic astronomy history
   - 1.3. Global description of the Milky Way: our present knowledge
2. Galactic Dynamics I: basic concepts
   - 2.1. Gravitational potentials & Poisso[illegible] TERESA *17, 18, 19, 25 sept*
   - 2.2. Orbits TERESA *26 30 set 1,2 oc*
   - 2.3. Collisionless dynamics MERCE *,3 [illegible]*

+2h hands-on orbits MERCE, *14 15 oct*

3. Astronomical measurements FEDE
   , 16 17 21 oct
   - 3.1. Astrometry
   - 3.2. Photometry
   - 3.3. Spectroscopy

+2h Exercises measurements FEDE *22 23 oct*

4. Statistical astronomy FEDE
   24,28 29 30 oct
   - 4.1. Apparent distribution of stars
   - 4.2. Stellar statistics fundamental equation
   - 4.3. Stellar luminosity function
   - 4.4. Initial Mass Function and Star formation History
   - 4.5. Galactic models for star count predictions
5. Galactic kinematics TERESA
   31 oct 4 5 6 nov
   - 5.1. Galactocentric reference systems
   - 5.2. Kinematics of solar neighbourhood stars
   - 5.3. Large scale kinematics
   - 5.4. Rotation curve and Oort constants
6. Galactic dynamics II: advanced
   - 6.1. Dynamics of spiral structure and bars *7 11 12 13 14 nov* MERCE
   - 6.2. Collisions and encounters of stellar systems MARK/INVITED *18 19 20 nov*
   - 6.3. Galaxy interactions, non-axisymmetry, Gaia TERESA *21,25 nov*

+2h hands-on kinematisc/Gaia recent data TERESA *26 27 nov*

+1h Journal Club TERESA *28 nov*

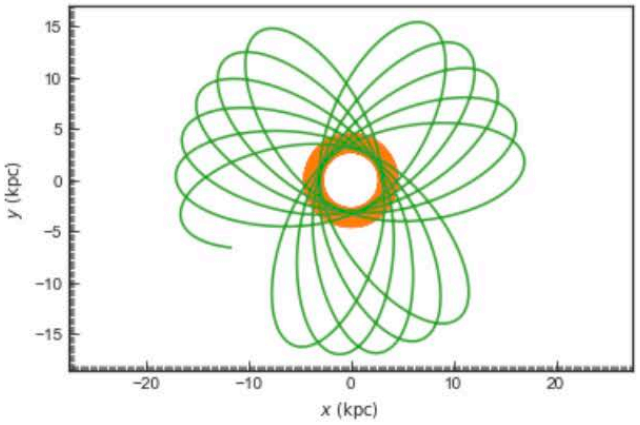
7. Chemical evolution of the Milky Way FEDE
   9 10 11 12 16 17 18 19 dec
   - 7.1. Observational evidence
   - 7.2. Surface gas density, supernova explosion rate and metal enrichment
   - 7.3. Basic elements of a model of chemical evolution
   - 7.4. Some simple models

Not 100% definitive

- **Hand-on exercises**
  - Analysis of *Gaia* data
  - Orbits in Galactic potentials

In [25]:
```
omw1.plot(d1='x',d2='y')
omw2.plot(d1='x',d2='y',overplot=True)
omw3.plot(d1='x',d2='y',overplot=True)

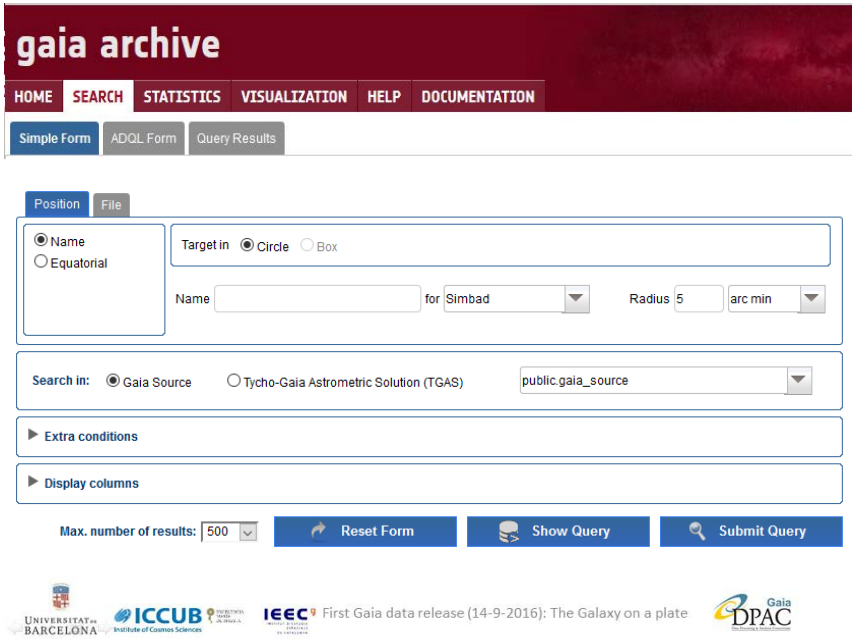
plt.axis('equal')
plt.show()
```



- **Paper reading and discussion (journal club)**

- **Short tasks TBD**

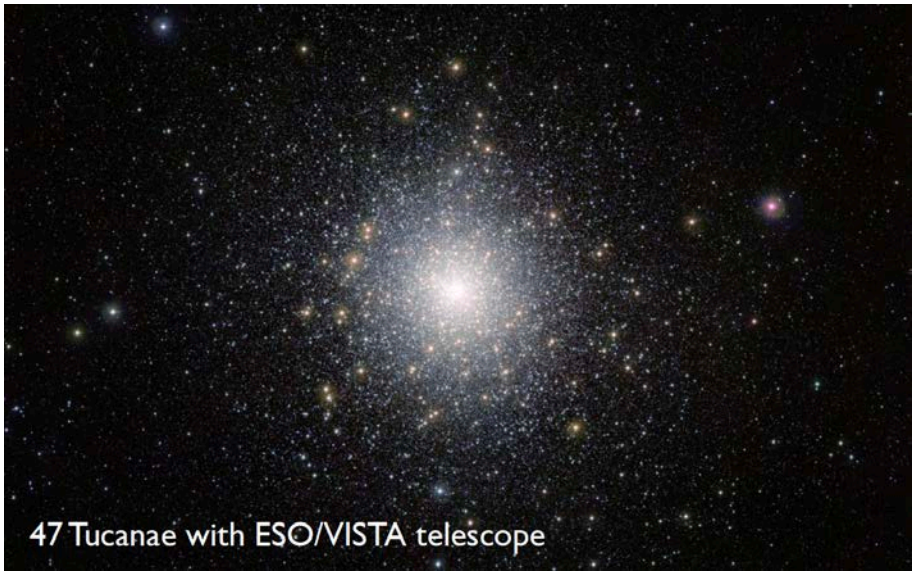
1. TBD: Gaia Archive & Gaia measurements / Statistical astronomy, Besançon model



HTTP://ARCHIVES.ESAC.ESA.INT/GAIA/

- **Lectures from invited professors**

TBC



Collisional dynamics

- Basic concepts for collisional systems: relaxation, core collapse, etc
- Dynamics of Globular Clusters

Nov 2023

Evaluation

Short tasks + presentations
Hands-on work
Participation
40%

Exam

5/10 required

60%

Exam date: TBD

https://campusvirtual.ub.edu/

- Detailed calendar
- Pdfs of lectures
- Material for hands-on, exercises, etc
- Additional material, papers, etc

Master's thesis

- Open clusters
- Galactic Disk dynamics
- Satellite galaxies
- Stellar evolution
- Population synthesis
- White dwarfs

- Big Data and Machine learning
- Galaxy formation and cosmology
- Spectroscopic stellar surveys
- Globular clusters
- Light pollution
- Chemical evolution of the MW

F. Anders, T. Antoja, L. Balaguer, J. M. Carrasco, F. Figueras, M. Gieles, X. Luri, E. Masana, M. Romero-Gomez, M. Semcsuk





